

QUEENSTOWN, South Africa

WMO: 686470

Lat: 31.918S

Lon: 26.878E

Elev: 3622

StdP: 12.87

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.3	33.0	10.3	10.6	58.9	14.1	12.8	59.5	25.9	60.8	22.3	58.9	1.8	90	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	92.3	62.9	89.1	62.6	86.1	62.2	69.1	80.1	67.6	78.7	66.4	77.3	10.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	109.3	73.4	64.5	104.2	71.9	63.2	99.5	70.7	35.6	80.1	34.3	78.9	33.2	77.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 21.1	18.5	16.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	70.8	70.4	67.7	62.0	57.2	51.7	51.3	55.2	59.6	63.1	65.7	69.3
	DBStd	9.39	6.04	5.46	6.23	6.02	6.23	5.96	6.55	7.19	7.56	7.52	7.01	6.27
	HDD50	171	0	0	0	4	13	50	63	28	8	4	1	0
	HDD65	2027	21	16	45	123	252	400	425	314	195	127	78	31
	CDD50	4532	645	571	550	363	235	100	103	190	296	410	471	598
	CDD65	916	202	167	130	33	9	1	1	12	33	67	97	164
	CDH74	7471	1580	1083	871	210	99	4	0	102	481	747	877	1417
	CDH80	2641	638	378	278	27	12	0	0	14	129	263	302	600
Wind	WSAvg	6.1	6.2	5.9	5.6	4.6	4.3	6.5	5.8	6.3	6.7	6.8	7.1	7.1
Precipitation	PrecAvg	19.4	2.5	3.2	2.9	1.5	0.6	0.6	0.4	0.7	0.9	2.0	2.3	2.6
	PrecMax	28.7	7.6	7.4	8.3	4.3	1.8	1.9	2.5	5.9	4.7	5.2	5.6	7.0
	PrecMin	12.0	0.0	0.3	0.3	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2
	PrecStd	5.0	1.7	2.0	1.9	0.9	0.6	0.6	0.6	1.2	1.0	1.5	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.0	93.9	91.9	84.3	81.1	73.2	72.7	82.0	88.7	91.9	92.1	97.1
		MCWB	64.3	65.0	66.0	61.3	57.7	51.7	48.9	53.8	56.9	58.3	61.2	62.5
	2%	DB	90.9	89.1	86.6	79.7	76.6	68.9	70.2	76.6	83.4	87.1	87.4	91.2
		MCWB	64.4	65.6	63.6	59.9	54.8	49.5	48.8	51.6	54.7	58.3	61.1	62.2
	5%	DB	87.0	84.8	82.8	76.5	73.3	66.4	68.0	73.1	79.4	82.6	83.3	86.6
		MCWB	64.3	65.2	63.4	58.6	53.7	48.0	47.8	50.6	53.7	57.4	60.2	62.7
	10%	DB	82.7	80.7	78.8	73.0	69.9	63.6	65.1	69.7	75.2	77.7	79.1	81.8
		MCWB	63.9	65.7	63.2	58.0	52.5	47.2	46.8	49.6	53.1	56.8	60.1	62.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	71.6	71.0	64.2	59.4	53.9	51.6	57.1	60.3	63.7	66.6	69.1
		MCDB	83.4	82.9	80.6	76.3	74.1	63.5	67.8	71.3	75.6	78.5	79.0	81.3
	2%	WB	68.1	69.3	67.9	62.5	56.9	52.2	50.1	54.6	58.4	61.8	64.8	66.8
		MCDB	80.7	79.2	77.6	73.0	72.0	62.4	66.5	68.6	74.9	76.4	78.1	79.2
	5%	WB	66.8	67.9	65.9	61.1	55.4	50.9	49.0	52.8	56.6	60.3	63.5	65.5
		MCDB	78.8	78.2	75.8	70.9	70.1	61.1	64.5	68.4	71.5	74.4	76.3	78.0
	10%	WB	65.8	66.5	64.3	59.8	53.9	49.5	47.8	51.1	54.8	59.1	62.1	64.2
		MCDB	77.1	76.1	74.5	69.3	66.1	59.9	62.1	65.3	69.5	71.7	73.8	76.7
Mean Daily Temperature Range	5% DB	MDBR	21.8	19.9	20.6	20.2	22.5	21.3	23.4	22.4	24.5	23.5	22.4	23.2
		MCDBR	29.2	26.8	26.5	25.6	27.4	25.0	25.6	27.6	31.6	32.6	30.5	32.0
		MCWBR	9.1	7.7	8.6	8.7	10.8	11.6	11.3	10.3	9.9	9.5	8.1	8.2
	5% WB	MCDBR	23.0	21.8	21.5	21.1	24.0	20.0	23.6	23.4	26.2	25.1	24.9	24.9
MCWBR		8.3	8.8	9.3	8.7	10.5	9.9	10.9	9.3	9.0	8.8	8.7	8.7	
Clear-Sky Solar Irradiance	taub	0.339	0.341	0.326	0.298	0.280	0.265	0.266	0.312	0.342	0.350	0.329	0.333	
	taud	2.508	2.501	2.514	2.571	2.590	2.597	2.583	2.435	2.344	2.367	2.481	2.516	
	Ebn at Noon	317	310	304	297	288	285	290	287	293	304	318	320	
	Edh at Noon	36	35	32	27	24	22	23	31	38	40	37	36	
All-Sky Solar Radiation	RadAvg	2218	2022	1730	1379	1112	987	1073	1317	1677	1962	2221	2333	
	RadStd	129	127	111	72	47	57	58	85	102	123	121	146	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page